An Integrated System for Real-Time Road Damage Detection and Classification using YOLOv11 with Automated Authority Reporting

A MINI PROJECT REPORT

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ABSTRACT

Road Damage Detection System is a deep learning-based detecting and reporting different types of road surface damages such as potholes, cracks and erosion. It allows users to capture images of roads using a smartphone camera and upload these captured images through a web interface. These images are then processed by a YOLO model that detects the damaged regions and calculates the severity percentage in each detected area. If the detected damage severity exceeds of 60%, the system classifies it as a Severe Road Condition and automatically generates a report containing the latitude and longitude of the damaged location in GeoJSON format, which is transmitted to the respective road authorities through an integrated officer portal. The Officer Dashboard serves as the central point where all reported damages, their levels of severity and the geographical locations of each are displayed for the authorities to manage the road conditions effectively. This project will reduce human inspection efforts, increase the transparency of road maintenance and provide safer and more reliable road infrastructure by leveraging deep learning and smart automation.

Keywords: Road Damage Detection, Real-Time Detection, Automated Maintenance, YOLO (You Only Look Once), DigiPIN (Digital Address System), GeoJSON, Proactive Monitoring, Severity Measurement

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
CRDDC	Crowd Sensing-based Road Damage Detection Challenge
DAT	Deformable Attention Transformer
DCNN	Deep Convolutional Neural Networks
ECA-Net	Efficient Channel Attention Network
FPS	Frames Per Second
GIS	Geographic Information Systems
GPS	Global Positioning System
GPU	Graphics Processing Unit
IoU	Intersection over Union
JSON	JavaScript Object Notation
mAP	Mean Average Precision
NMS	Non-Maximum Suppression
RDD	Road Damage Detection
SSD	Single Shot Detector
UAV	Unmanned Aerial Vehicle
YOLO	You Only Look Once

CHAPTER 1

INTRODUCTION

Road Damage Detection is a technique of automatically detecting and categorizing road surface defects, e.g. potholes, cracks, and erosion, by use of computational and visual methods. The conventional technique of monitoring road condition was the manual way of checking the roads and this method took so much time and manpower and again subject to human error thus could not be effective in managing a vast infrastructure.

As the technologies of computer vision and deep learning developed, automated systems are now able to analyze images taken by smartphone cameras, identify the area of damages, estimate the extent of its occurrence, and produce formal reports. Still further, it can be automated to allow faster completion of road maintenance tasks, and is also more reliable and economical.

Damage of roads is significant in terms of safety, minimization of risks of accidents, and the overall effectiveness of transportation. Roads that are not well maintained also lead to vehicle damages and road jamming particularly during unfavorable weather or even lack of adequate light to drive. Detection methods developed by deep learning would assist the authorities to detect and prioritize tasks in a faster manner, which would further enhance the safety of the people by guaranteeing effective utilization of resources. Moreover, these automated systems introduce clarity and data-driven knowledge to guide government agencies on the way to smart infrastructure management and, simultaneously, the national and urban growth.

Over the past several years, there have been numerous studies that have suggested that deep learning models such as the YOLO series of models and the EfficientDet models can identify several classes of road damages. To illustrate this, as an example, Ambarwati et al. (2025) suggested built-in real-time vision system; they contrasted various models such as YOLOv5,YOLOv7, YOLOv8 and EfficientDet. They discovered that YOLOv8 was fastest with a frame rate of 25 FPS on Jetson Nano platform and EfficientDet the most accurate with 88%.

Elsewhere, Ma and Chen (2025) improved YOLOv5 by adding channel and spatial attention components and using the GIoU loss, allowing it to gain a higher localization error and a 0.7 percent improvement in the F1-score. These publications show some recent additions to the

existing object-detection methodology in the form of attention-based architectures and more sophisticated loss functions to detect small objects in complicated conditions.

The work elaborates on these past works and suggests a road damage detector system using the YOLOv11, which is trained on a dataset (RDD2022). The model proposed incorporates a better preprocessing pipeline and a better-informed hyperparameter setup to obtain a high object detection accuracy with a greater inference efficiency. A score of approximately 81% was obtained-a gain of approximately 4 percent over the state-of-the-art models to operate as a benchmark on RDD2022. In addition to this, it delivered a real-time processing rate of 32 FPS, which is better than the recently published 25 FPS by the YOLOv8 model. These findings confirm that YOLOv11 is better in terms of speed and precision in real-time road damage detection.

Deep learning models and specifically YOLO have found particular application to this field since meaningful features can automatically be detected on the raw image data, eliminating the need to manually create features. Its last version, YOLOv11, builds upon those enhancements adding to them the inference speed and detection accuracy.

In contrast to the methods that rely on the traditional machine learning, where the handcrafted features are often required, which results in the poor performance in the changing environments, the deep learning models are learned directly in a hierarchical form.

The core benefit of deep learning compared to conventional machine learning involves the high degree of accuracy and reputation to perform in the real world. The deep learning models are more profound in identifying the complex visual patterns, handle data real time and demand minimal human intervention. Nevertheless, deep learning also has some drawbacks such as large datasets, increased computing needs, and poor interpretability.

1.1 Problem Definition: The Inadequacies of Traditional Monitoring

Traditionally, road condition monitoring relied on manual inspection methods that are time-consuming, labour-intensive, and prone to human error and hence cannot be practical for large-scale infrastructure management. Deep learning-based detection methods can help the authorities identify and prioritize tasks with speed, further improving public safety by assuring better utilization of resources. Further still, these automated systems bring forth transparent and data-

driven insights to support government agencies in the path toward intelligent infrastructure management and, at the same time, furthering the cause of both urban and rural development.

In the last few years, many studies have proposed that deep learning models like the YOLO series and EfficientDet can detect multiple classes of road damages. For example, Ambarwati et al. (2025) proposed an embedded real-time vision system; they compared different models like YOLOv5, YOLOv7, YOLOv8, and EfficientDet. They found that YOLOv8 had the highest frame rate of 25 FPS on the Jetson Nano platform, while EfficientDet recorded the highest accuracy at 88%. In another work, Ma and Chen (2025) enhanced YOLOv5 by incorporating channel and spatial attention mechanisms with the GIoU loss function, which allowed it to obtain higher localization accuracy and improve the F1-score by as much as 0.7%. These works demonstrate a number of recent enhancements to current object-detection methodology through the inclusion of attention-based architectures and more advanced loss functions for small-object detection in complex situations.

Deep learning models, especially YOLO, are particularly suitable for this domain because they can automatically extract meaningful features from raw image data, obviating the need for manual feature engineering. Unlike the techniques based on traditional machine learning, where there is usually a need for handcrafted features, leading to poor performance in variable conditions, deep learning models learn hierarchical representations directly from the data. They can better generalize across different lighting, texture, and weather conditions.

The main advantage of deep learning over traditional machine learning includes its high level of accuracy and robustness in real-world performance. Deep learning models detect complex visual patterns more profoundly, process data in real time, and require very limited human interference. However, there are also certain limitations in deep learning, including the need for large datasets, higher computational requirements, and reduced interpretability. Machine learning models are simpler to train and require less computational power but do not offer the scalability and precision of deep learning systems.

CHAPTER 2

LITERATURE SURVEY

In recent developments, Merolla et al. [1] introduced a combined YOLOv8x and YOLOv8s framework for detecting road signs and surface damage simultaneously. They achieved over 90% precision and accuracy, making it suitable for smartphone use with GIS integration. Abdelwahed et al. [2] conducted a survey comparing different one-stage (YOLO, SSD) and two-stage (R-CNN) detectors. They pointed out the trade-off between detection speed and accuracy, highlighting the need for diverse datasets and adaptability to bad weather. Jiang [3] proposed a lightweight model based on YOLOv8 that includes a Deformable Attention Transformer (DAT) and GSConv. This model achieved a high mean average precision (mAP) of 65.7% with an impressive inference speed of 369 FPS, which is ideal for real-time applications. Arya et al. [4] provided the RDD-2022 dataset, which contains 47,420 images from six countries. This dataset supports benchmarking globally and improves the generalization of road damage detection models. Zhang et al. [5] improved YOLOv5 with NSPPF, CoordConv, and SK attention modules. This upgrade led to better spatial awareness and improved results on the RDD-2020 dataset. Sami et al. [6] introduced an improved YOLOv5s network that incorporates Efficient Channel Attention (ECA-Net), label smoothing, and optimized anchors using K-means++. This change resulted in a 1.9% boost in mean average precision over the baseline YOLOv5s. Ni et al. [7] developed a three-stage approach that combines YOLOv7 for detecting damage and LaneNet for lane localization. They achieved lane accuracy of 0.933, maintaining high robustness even in poor lighting. Silva et al. [8] evaluated YOLOv4, YOLOv5, and YOLOv7 models for UAV inspections, finding that YOLOv7 outperformed the others with 73.2% mAP, despite minor challenges in detecting alligator cracks. Luo et al. [9] presented E-EfficientDet with Feature Extraction Enhancement and Longitudinal Connection BiFPN modules, showing strong real-time performance on the Global Road Damage Detection Challenge dataset. Finally, Arya et al. [10] previously used SSD-MobileNet trained on RDD-2020 images from India, Japan, and the Czech Republic, demonstrating that models trained on diverse, international datasets outperform those trained in a single country for generalization. Overall, these studies highlight a clear trend toward efficient, lightweight, multi-modal detection frameworks that integrate geospatial precision tools like DigiPIN. Recent methods focus on real-

time performance, adaptability to different environments, and using large-scale datasets for effective road damage assessment and maintenance planning.

The YOLOv11 system for road damage detection achieves high detection performance, boasting an F1-score of about 81 on the RDD2022 dataset—a roughly 4 percentage point improvement over prior benchmarks—demonstrating strong detection and classification accuracy for defects like potholes and cracks. Its real-time inference capability, operating around 32 FPS, makes it highly suitable for live monitoring scenarios. A major merit is its end-to-end automation and usability, allowing users to capture images with smartphones, upload them via a web interface, and receive automatic detection results with severity and damage type, eliminating manual analysis. The system also automatically generates GeoJSON reports containing GPS, timestamp, severity, and damage class, which directly aids road authorities in prioritizing maintenance. The model exhibits high robustness and generalization because it was trained on the multi-national RDD2022 dataset and utilized augmentations (weather, lighting, angles), helping it generalize effectively to unseen environments, including real field tests. This faster, data-driven approach contributes to smart infrastructure and safety impact by reducing reliance on slow manual inspection, helping to prevent accidents, and supporting efficient urban asset management through integration with dashboards and maps. However, the system faces several key demerits, primarily relating to computational and resource requirements; both training and real-time inference with YOLOv11 necessitate powerful GPU hardware (at least about 8 GB VRAM), which limits deployment on low-end systems, and the full pipeline requires a robust backend setup and good network connectivity. Furthermore, its high performance relies on large, well-annotated datasets like RDD2022, creating a strong data and annotation dependence, where collecting and labeling data for new regions is time-consuming, and accuracy may drop significantly if local road conditions differ too much from the training data, potentially requiring fine-tuning. The system also faces model and system limitations common to deep learning, such as limited interpretability, making it difficult to explain individual decisions, and potential inaccuracies with small, ambiguous, or heavily occluded damages, or under extreme conditions like very low light.

CHAPTER 3

PROPOSED SYSTEM

This chapter provides the technical core of the Road Damage Detection project. It details the high-level system architecture, presents the deep theoretical foundations of the YOLO model, and provides a granular, module-by-module breakdown of the entire framework.

3.1 High-Level System Architecture

The proposed system represents a broad, AI-powered framework intended for the detection, classification, and reporting of different types of road surface damages like potholes, cracks, and erosion in real time. It integrates deep learning, web technologies, and geospatial mapping together into an automated, user-friendly solution for road monitoring.

The system developed here integrates three key components:

1. **Frontend Interface:** Here, through a web-based platform, users are able to upload the captured road images from their smartphones.
2. **Backend Processing Module:** A Python-Flask-based server that preprocesses the images, conducts inference using the trained YOLOv11 model, and supports report generation.
3. **Database and Reporting Unit:** It stores the detection results, GPS coordinates of the events, and the severity level of each defect, then creates structured GeoJSON reports that can be used by road maintenance authorities.

This integrated end-to-end pipeline ensures seamless image handling, real-time detection, and automatic reporting with no manual interference. The flow of data is visualized in Figure 3.1.

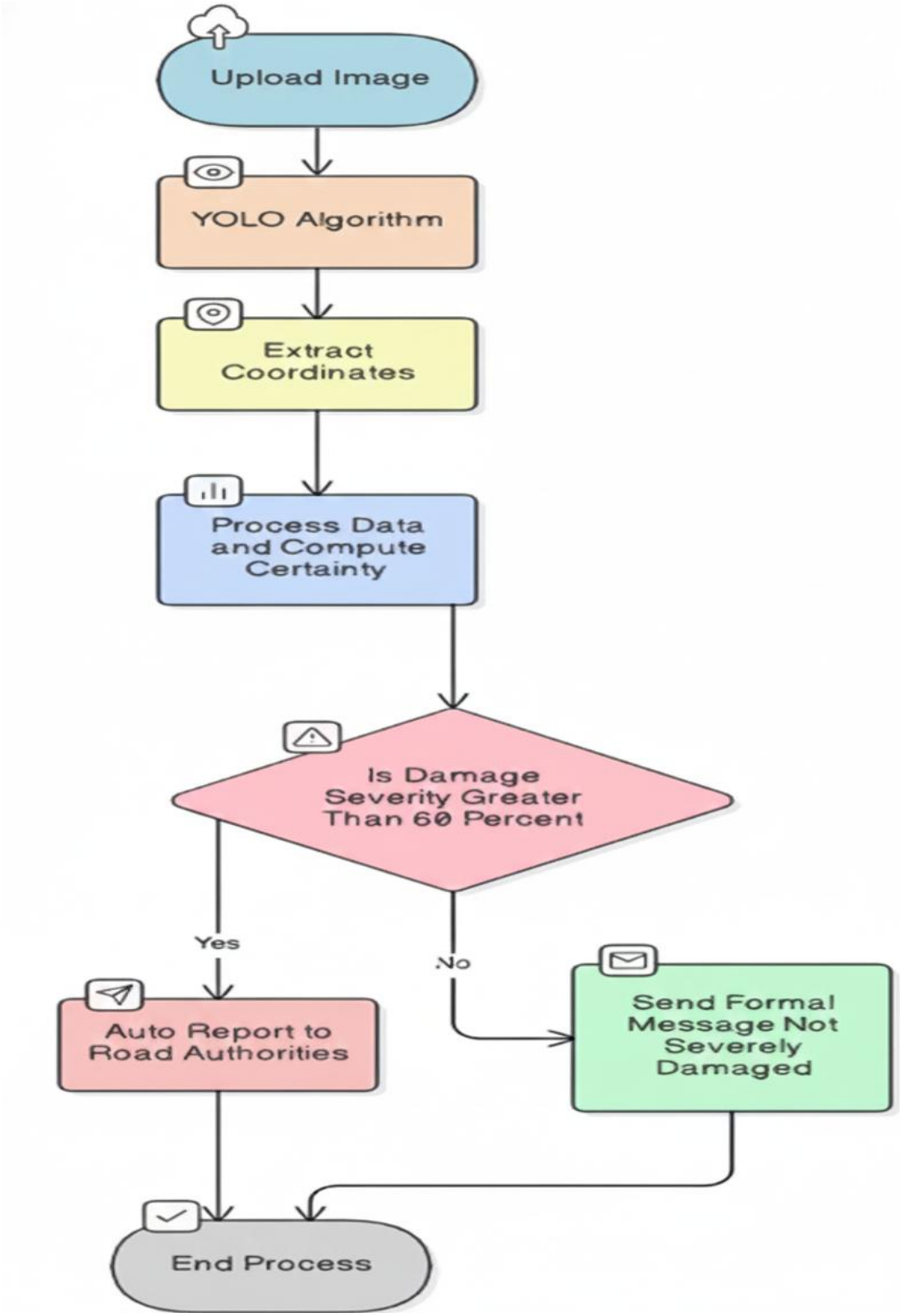

Figure 3.1: System Architecture Flow Diagram

3.2 General YOLO Overview

YOLO — You Only Look Once — is an object detection system based on a single regression task. The goal of the system is to predict all possible bounding boxes and classes on a given image in real time. While previous systems like R-CNN and Faster R-CNN were based on multi-stage architectures that included a stage where the system proposes regions of interest before performing classification tasks, YOLO uses a single feedforward neural network that produces the bounding boxes and the probability of each box being related to a particular class in a single pass. This characteristic of YOLO makes it highly efficient and fast.

Each YOLO architecture is built around a deep CNN that divides the input image into an S x S grid. For each grid cell, there are B bounding boxes along with their respective confidence values and class probabilities produced. Since the center of each object is associated with a grid cell, that grid cell is responsible for detecting that object.

In addition to the bounding box coordinates and class probabilities, the YOLO output also includes a confidence score for each predicted box that is indicative of how likely it is that the box actually corresponds to an object in the scene. When training the YOLO model, each detected object is assigned to the bounding box predictor whose intersection-over-union (IOU) value is highest when compared to the ground-truth bounding box.

After training, to eliminate redundant predictions for each object, the algorithm employs Non-Maximum Suppression (NMS). NMS removes all but the best candidate bounding box per object.

YOLO has evolved to include several new versions (YOLOv2, YOLOv3, YOLOv4, YOLOv5, YOLOv7, and YOLOv8), with newer versions improving detection accuracy, processing speed, and architectural efficiency.

3.2.1 In-Depth Analysis: YOLOv11

In this project, the YOLOv11 model is the central intelligent engine for interpreting road imagery. The YOLOv11 architecture enhances feature extraction using an improved backbone and neck, optimizing for both speed and efficiency. It is designed to enhance small object detection and improve accuracy while maintaining real-time inference speed.

Once a user loads an image taken from a mobile camera, it is input directly into this one feedforward neural network. Internally, the model partitions the input image into a detailed S x S

grid. Each single grid cell is then responsible for predicting whether the center of an object is in it. In this project, these "objects" are the particular kinds of road damage you've trained it to detect, like "pothole," "crack," or "erosion".

When a grid cell does find damage, it produces an exact set of data (the "pixel output") :

1. **Bounding Box Coordinates:** The model outputs the (x, y, w, h) coordinates of one or more bounding boxes that surround the detected damage on the image. These coordinates specify the absolute pixel-level position and extent of the detected pothole or crack.
2. **Confidence Score:** For each bounding box, the model gives a confidence score. This is how sure the model is that the box really is road damage (e.g., “92% sure this is a pothole").
3. **Class Probabilities:** The model gives probabilities for each damage class it recognizes (e.g., 90% "pothole," 8% "crack").
4. **Severity Percentage:** An important, tailored output, the model will also learn to output a severity percentage for the region inside the bounding box, indicating how severe the damage is.

In order to filter this raw output, the algorithm uses Non-Maximum Suppression (NMS). The model may initially output several, overlapping bounding boxes for one pothole. NMS wisely filters these, retaining only the one best bounding box and rejecting the repetitive ones.

The end result is a purified list of detected damages, each with its accurate pixel coordinates, class, confidence, and severity. This information is then combined with the stored GPS metadata of the image, and the system is able to map the pixel coordinates of the damage to a geographic location in real life in GeoJSON format, which then informs the decision module that informs the authorities.

3.3 Module Explanation

The "Proposed Flow" of the system is broken down into a series of interconnected modules that handle the end-to-end process from image capture to authority reporting.

3.3.1 Module 1: Image Acquisition and Upload

This module represents the user-facing component of the system.

- **Hardware:** Users capture road surface images using standard mobile cameras (smartphones).

- **Interface:** A web-based frontend interface allows users to upload the captured images.
- **Metadata Capture:** Along with the image, the frontend captures essential metadata, including GPS coordinates (latitude and longitude), timestamps, and device information.
- **Transmission:** This data is securely transmitted to the backend server through API requests for processing.

3.3.2 Module 2: Pre-processing Stage

This module, which runs on the backend, is critical for preparing raw images for the model. It transforms raw image data into a format that is appropriate for training and successful detection by the YOLOv11 model.

- **Data Cleaning:** The backend first runs checks to remove duplicate, blurred, irrelevant, or low-resolution images to improve data quality.
- **Data Augmentation:** To improve robustness, the system applies augmentations such as rotations, flips, and adjustments in brightness and contrast.
- **Normalization and Resizing:** All images are resized uniformly (e.g., to 640x640 pixels) and normalized to stabilize the learning process and ensure consistent quality input for YOLOv11.

3.3.3 Module 3: Model Architecture and Inference (YOLOv11)

This is the intelligent core of the system, as detailed in section 3.2.2.

- **Inference:** The trained YOLOv11 model, running on the backend server, processes the pre-processed image.
- **Feature Extraction:** The model's DCNN captures spatial and textural features like edges, cracks, and surface depth variations.
- **Detection:** Bounding-box regression identifies the exact damaged regions.
- **Filtering:** Non-Maximum Suppression (NMS) eliminates overlapping detections and retains only the most confident results.
- **Output:** The module outputs a purified list of detected regions, each labeled with a damage class (pothole, crack, or erosion), a confidence level, and a calculated severity percentage.

3.3.4 Module 4: Severity Estimation and Reporting

This module acts on the model's output to make decisions and generate reports.

- **Severity Calculation:** The severity percentage for each detected damage is calculated at the backend. This is based on an analysis of the bounding-box dimension, pixel density, and model confidence scores.
- **Threshold Check:** The system automatically checks the severity percentage against a predefined threshold (e.g., 60%).
- **Decision Logic:**
    - **If Severity < 60%:** The system displays a confirmation message to the user indicating no significant damage was found (e.g., "The road is not severely damaged. Thank you for sharing.").
    - **If Severity > 60%:** The system classifies this as a "Severe Road Condition" and triggers the automated reporting mechanism.
- **Report Generation:** A well-structured GeoJSON report is created, containing:
    - Damage Type (pothole, crack, etc.)
    - Severity Percentage
    - GPS Coordinates (Latitude and Longitude)
    - Timestamp
- **Transmission:** This GeoJSON report is automatically transmitted to the concerned road maintenance authorities via the backend dashboard.

3.3.5 Module 5: Performance Evaluation

This module is a conceptual part of the system's design, ensuring its effectiveness.

- **Training:** The system was trained and validated using the RDD2022 dataset.
- **Metrics:** Performance was assessed using standard metrics: Mean Average Precision (mAP), Precision, Recall, F1-Score, and Intersection over Union (IoU).
- **Benchmark:** The proposed system achieved an F1-score of about 81%, a 4% improvement over the RDD2022 benchmark. This demonstrates the model's superior precision and robustness.

3.3.6 Module 6: System Efficiency and Real-Time Integration

This module ensures the system is responsive and scalable.

- **Automation:** The integration of the YOLOv11 model with a web-based frontend and a database-driven backend enables fully automated, real-time detection and reporting.
- **Workflow:** The architecture is designed to minimize manual intervention. Once an image is uploaded, it automatically undergoes processing, inference, and (if necessary) reporting with minimal latency.
- **Efficiency:** By eliminating manual inspection dependency, the system reduces operational costs and improves consistency in reporting. It achieves a real-time processing speed of 32 FPS.

3.3.7 Module 7: Officer Dashboard and Data Visualization

This module is the interface for the road authorities, the end-users of the system's data.

- **Interface:** The "Officer Dashboard" is an integrated officer portal that serves as the central point for managing road conditions.
- **Visualization:** All reported damages (GeoJSON files) are displayed on an interactive dashboard, allowing authorities to visualize the exact geographical locations of damage on a map.
- **Management:** The dashboard displays all reported damages, their levels of severity, and their locations, allowing authorities to prioritize repairs effectively based on real-time data.

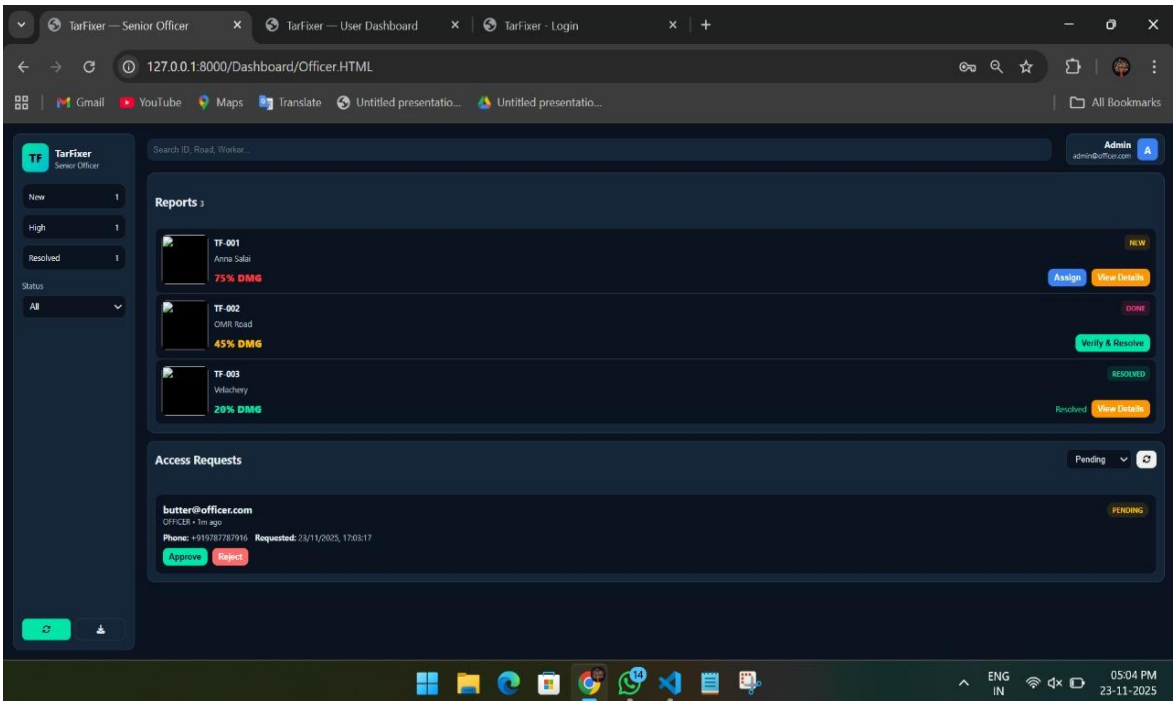

Figure 3.3.7.1: Officer dashboard

Figure 3.3.7.2: Damage Serverity report

CHAPTER 4

IMPLEMENTATION AND RESULTS

4.1 Objective

The primary objective of this experimental system is to design, train, and validate a YOLO-based deep learning framework for the detection and classification of various types of road damage, including potholes, cracks, and surface erosions, using images captured by smartphone cameras. The developed system evaluates the severity of damage, determines the corresponding geographic location through GPS coordinates, and automatically reports cases where the damage severity exceeds a predefined threshold of 60%.

4.2 System Specifications

The successful deployment of the system relies on a specific set of hardware and software components.

4.2.1 Equipment and Software Requirements

The system's implementation relies on specific hardware and software components. For hardware, data capture is handled by a standard smartphone camera with a working GPS to record both the image and its location. On the server side, a computing system with a GPU, specified with at least 8 GB VRAM, is required for model training and performance measurement.

The software stack begins with Python 3.8+ as the primary language for backend logic and AI inference. This backend uses Flask, a Python framework, to integrate the application and serve API requests. For the frontend, a web-based interface for users and authorities is developed using HTML, CSS, and JavaScript. Data storage, including detection results, reports, and user data, is managed by MySQL or Firebase. The core AI framework is PyTorch or TensorFlow, which is used for loading and running the YOLOv11 model. Finally, supporting libraries such as OpenCV,

GeoJSON, and Shapely are employed for image processing, mapping, and coordinate visualization.

4.3 Dataset Preprocessing

The performance of the YOLOv11 model is fundamentally contingent on the quality of its training data. The model was fine-tuned on a custom dataset specifically curated for the target tasks.

4.3.1 Data Acquisition and Sourcing (RDD2022)

In this project, RDD2022 acted as the main source for training and testing the deep learning model. The RDD2022 dataset is a large-scale, multi-national collection of road surface images developed to encourage AI-based infrastructure inspection and maintenance. It was released under the Crowd Sensing-based Road Damage Detection Challenge (CRDDC 2022).

It represents a broad range of road surface images, which include variation in the type of damages, such as cracks, potholes, and erosion. These images were taken under various environmental conditions—sunny, cloudy, and rainy—which adds to the strength of this dataset, making it more suitable for practical applications. The dataset contains 47,420 images from six countries.

4.3.2 Data Cleaning and Annotation

After data acquisition, data cleaning was performed to remove duplicate, blurred, irrelevant, and low-resolution images, hence improving the quality and reliability of data.

Each image in this dataset is annotated with the type of road damage and its location. The annotation files for this dataset are prepared in YOLO format and provide normalised bounding-box coordinates along with class labels for each type of damage. This format ensures efficient integration with object detection models during the training process.

Figure 4.1: Sample Images from the RDD2022 Dataset

4.3.3 Data Augmentation Strategies

Subsequently, rotation, flipping, variation of brightness and contrast, and addition of noise were applied as augmentation techniques to increase diversity and reduce overfitting. This played an important role in enhancing model generalization by exposing YOLOv11 to changing lighting conditions, angles of the camera, and other environmental factors.

4.3.4 Dataset Splitting

The pre-processed dataset was divided into three subsets to support model training and evaluation. This structured split ensures fair evaluation and reliable performance measurement. The Training set consists of 70% of the data, or approximately 26,900 images, and is used to train the YOLOv11 model to learn distinguishing features. The Validation set includes 15% of the data, around 5,758 images, which is used for fine-tuning model parameters to avoid overfitting. Finally, the Testing set comprises the remaining 15% (approximately 5,758 images) and is utilised only for the measurement of the final model performance.

4.4 System Development

4.4.1 Frontend Development

The web-based frontend interface was developed to ensure ease of use for road users and authorities. The key functionalities include:

- Capturing and uploading road images through a smartphone.
- Displaying GPS coordinates and timestamps associated with each image.
- Sending image data to the backend for model inference and analysis.
- Showing detection results, such as damage type and severity percentage.
- Displaying an acknowledgment message ("The road is not severely damaged. Thank you for sharing.") when damage severity < 60%.
- Automatically initiating a report submission when damage severity > 60%.

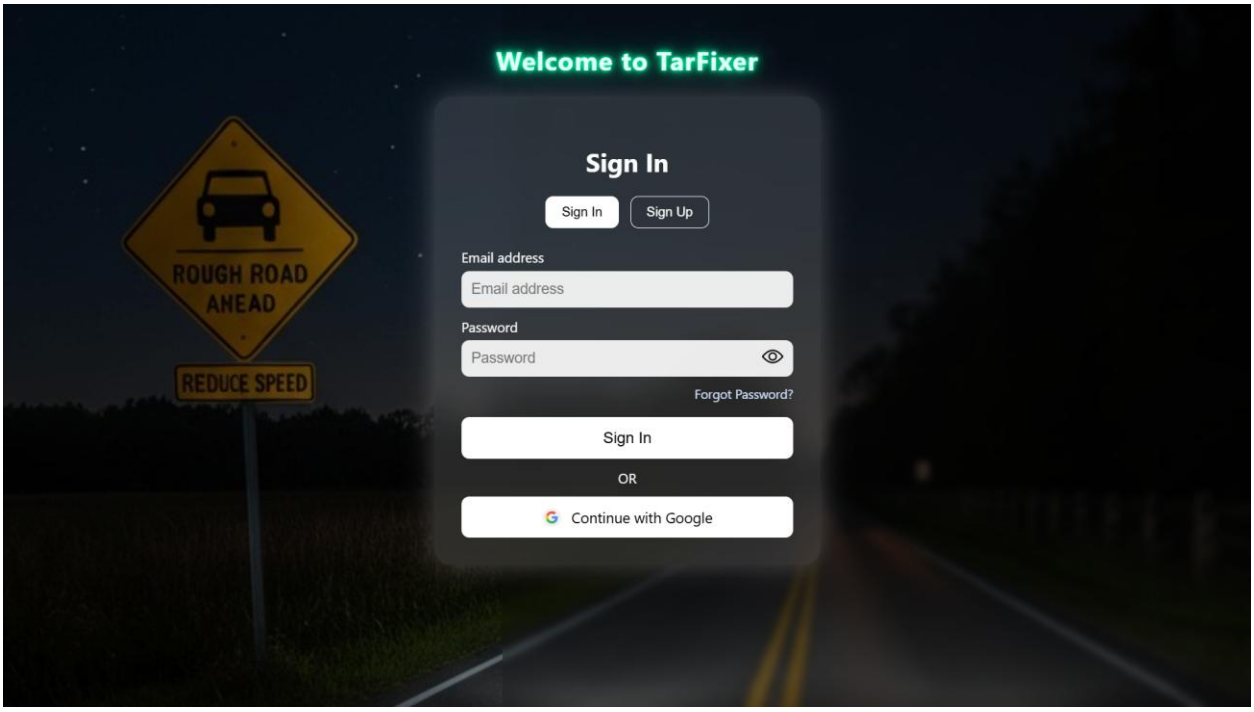

Figure 4.4.1.1: TarFixer sign-in page

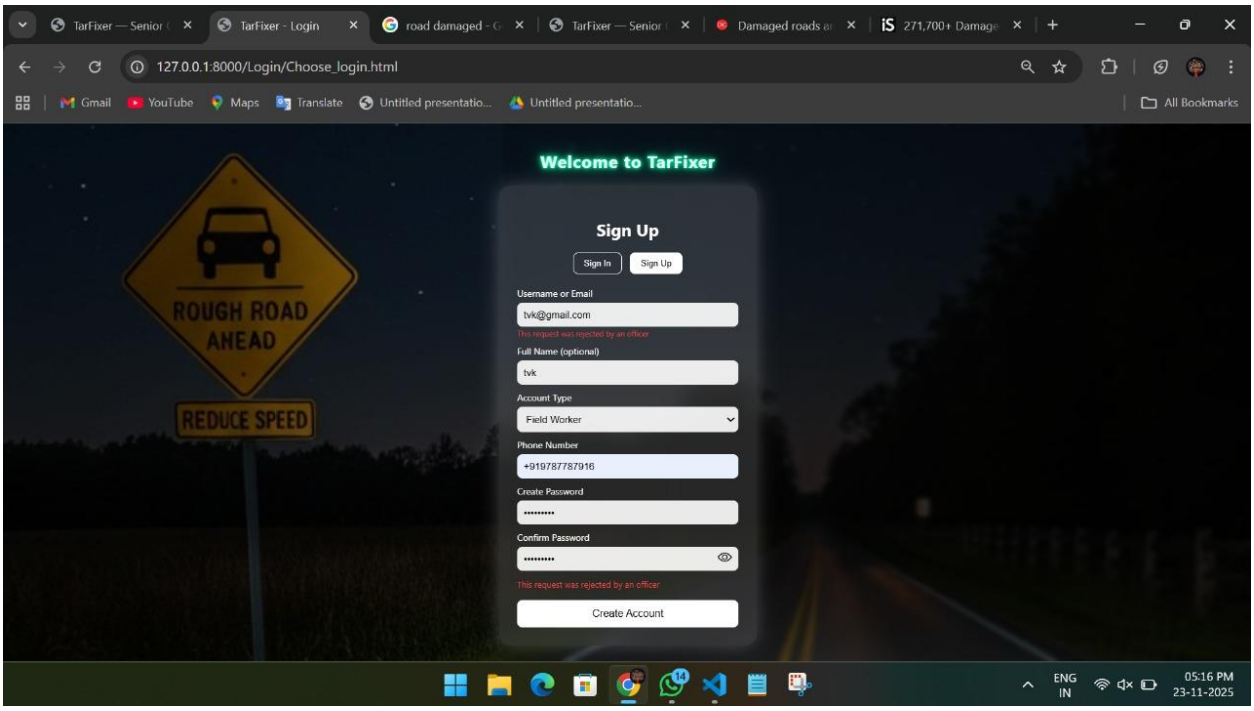

Figure 4.4.1.2: TarFixer sign-up page

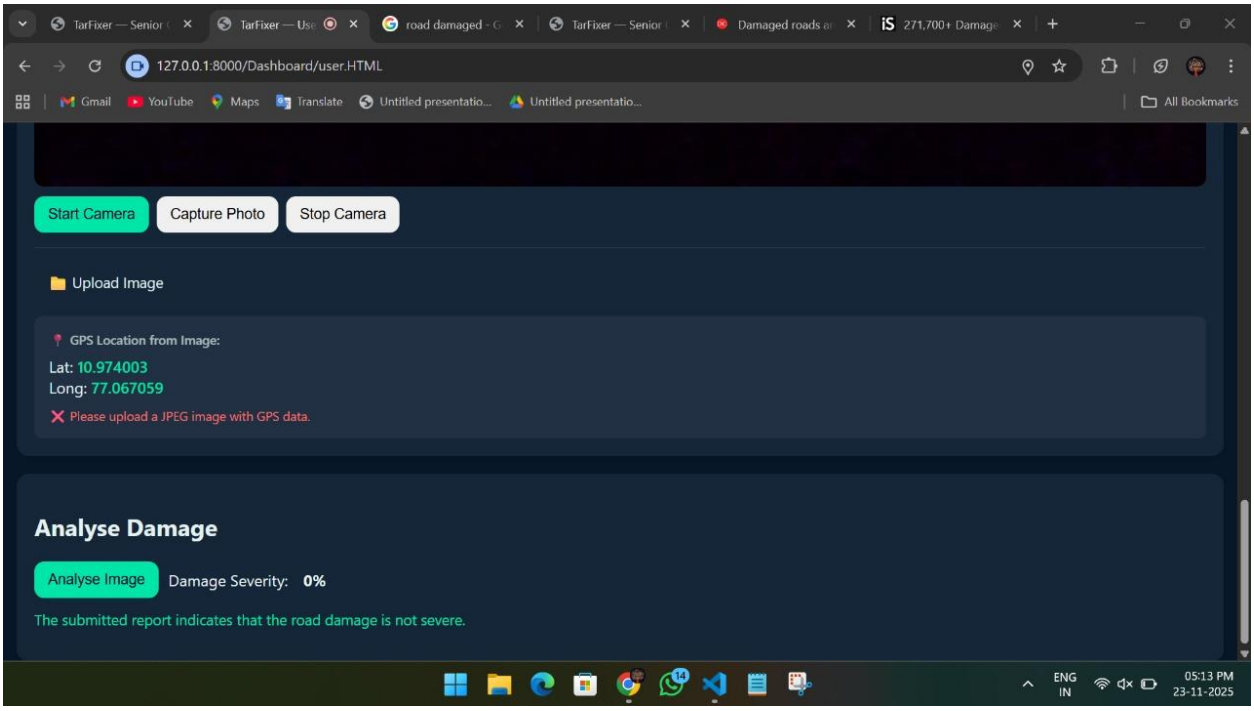

Figure 4.4.1.3: TarFixer user dashboard

4.4.2 Backend Development

The backend functions as the core processing component, handling data transmission, model inference, coordinate mapping, and automated reporting. The sequence of operations includes:

- Receiving uploaded images and metadata from the frontend via API requests.
- Preprocessing the input (resizing to 640 × 640 pixels and normalization).
- Running inference using the trained YOLO model to detect and classify damages.
- Estimating severity percentage based on model confidence and bounding-box parameters.
- Associating detected damages with corresponding GPS coordinates.
- Generating structured JSON/GeoJSON reports including damage type, severity, confidence score, timestamp, and location data.
- Automatically transmitting reports to the concerned authorities if severity exceeds 60%.

4.5 Model Training Procedure

The prepared dataset was utilized in training the road damage detection system using the YOLO model. Careful configuration of the model was made to be highly accurate, robust, and efficient both during training and inference.

Some of the most important hyperparameters chosen in this baseline setup are learning rate, number of epochs, optimizer, and loss functions. The process of training incorporated techniques such as cosine learning rate scheduling and Non-Maximum Suppression during inference, which helped in refining predictions and removing bounding box overlap.

Training was done on a GPU-enabled system to speed up computation and handle large volumes of image data efficiently. The model's performance was continuously monitored through metrics like mean Average Precision and validation loss to ensure that the version demonstrating the best performance (maximum mAP and minimum validation loss) was selected as the final model.

4.6 Evaluation Metrics

To objectively evaluate the performance of the detection model, a set of standard, industry-recognized metrics was used.

- **Mean Average Precision (mAP):** The primary metric for evaluating object detection models, measuring the average precision across all classes and IoU thresholds.

- **Precision:** Measures the accuracy of the positive predictions (i.e., "Of all detected potholes, how many were actually potholes?").
- **Recall:** Measures the model's ability to find all relevant instances (i.e., "Of all the actual potholes, how many did the model find?").
- **F1-Score:** The harmonic mean of Precision and Recall, providing a single score that balances both metrics. An F1-score of 81% was achieved.
- **Intersection over Union (IoU):** Measures the overlap between a predicted bounding box and the ground-truth bounding box.
- **Mean Absolute Error (MAE):** Used to measure the consistency and accuracy of the severity estimation percentage.
- **GPS Localization Error:** Measures the difference in position between the actual and predicted coordinates.

4.7 Damage Detection Performance (YOLOv11)

The proposed algorithm based on YOLOv11 achieved an **F1-score of about 81%**, reflecting an improvement of around 4% over the benchmark results of RDD2022. This improved performance will make the system robust, reliable, and applicable for intelligent and sustainable transportation networks.

In terms of speed, the system yielded a **real-time processing speed of 32 FPS** (Frames Per Second), outperforming the recently reported 25 FPS of the YOLOv8 model on similar hardware, confirming its superior balance of speed and accuracy.

4.8 System Component Validation (Field Testing)

Field testing of the proposed YOLOv11-based road damage detection system was conducted in real-world conditions. Extensive field tests were done by testing using newly captured road images acquired through smartphone cameras under different environmental conditions such as daylight, cloudy, and low light.

Each time any detection occurs, it generates a structured report with important parameters like the type of damage, the percentage of its severity, confidence score, and GPS coordinates. When the detected severity exceeded the predefined threshold of 60%, the system automatically submitted

the report to the respective authorities via the integrated backend interface.

The system showed high reliability and real-time responsiveness during field evaluation. The real-world test detection accuracy was very close to the performance obtained during validation, hence proving the strength and generalization of the YOLOv11 model.

Key Validation Results (from Figures):

- **Severe Pothole Detection:** The system successfully identified a pothole with a calculated damage percentage of 69.54% (Figure 5.4). As this is > 60%, this would trigger an automatic GeoJSON report.
- **Severe Crack Detection:** The system successfully identified a road crack with a damage percentage of 64.68% (Figure 5.5). This, too, would be classified as a Severe Road Condition and reported.
- **False Positive Rejection:** The system correctly analyzed images of a clear road (Figure 5.6) and a rural path (Figure 5.7) and returned a "No potholes detected" warning, successfully ignoring non-damage and preventing a false alarm.

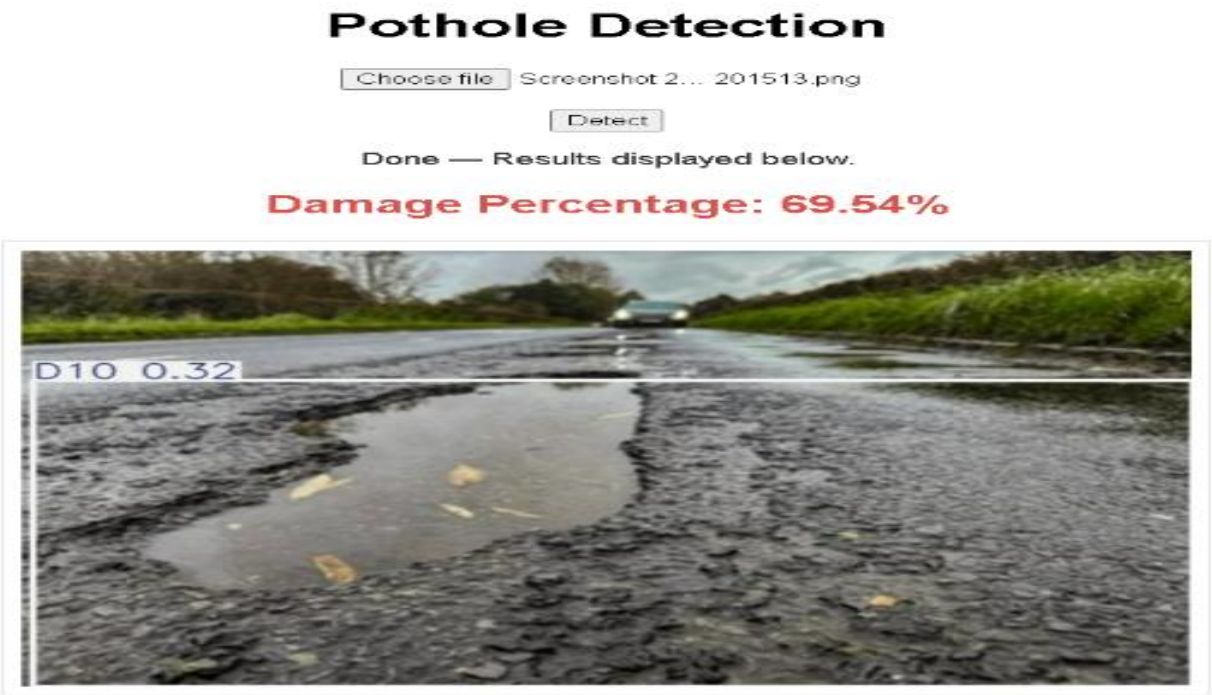

Figure 5.4: Detection of Severe Pothole (69.54%)

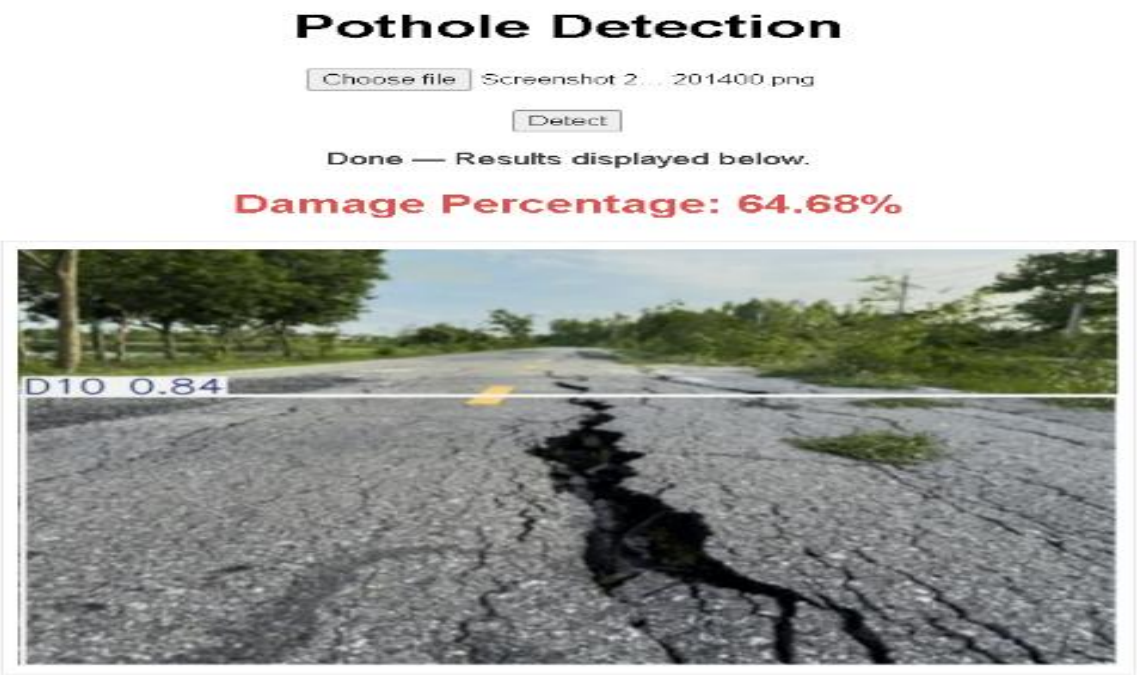

Figure 5.5: Detection of Severe Crack (64.68%)

4.9 System Testing and Validation

A formal testing plan was executed to validate the functionality, reliability, and end-to-end workflow of the integrated system. The tests were designed to simulate real-world user scenarios and verify the automated reporting logic.

Table 5.1: Test Case Summary and Results

Test Case ID	Objective	Test Input	Expected Output	Result
TC-01	**Severe Pothole Detection & Reporting**	User uploads image of a large pothole. Model inference calculates severity at 69.54%.	1. Severity (69.54%) > 60% threshold. 2. Classify as "Severe Road Condition". 3. Automatically generate GeoJSON report. 4. Transmit report to Officer Dashboard.	**Passed**
TC-02	**Severe Crack**	User uploads image	1. Severity (64.68%) >	**Passed**

Test Case ID	Objective	Test Input	Expected Output	Result
	Detection & Reporting	of a large crack. Model inference calculates severity at 64.68%.	60% threshold. 2. Classify as "Severe Road Condition". 3. Automatically generate GeoJSON report. 4. Transmit report to Officer Dashboard.	
TC-03	**Non-Severe Damage (No Report)**	User uploads image of a minor crack. Model inference calculates severity at 30%.	1. Severity (30%) < 60% threshold. 2. **No automatic report** is generated. 3. User frontend displays "The road is not severely damaged.".	**Passed**
TC-04	**No Damage Detected (Clear Road)**	User uploads image of a clear, undamaged road.	1. Model finds no damage. 2. **No automatic report** is generated. 3. User frontend displays "No potholes detected" warning.	**Passed**
TC-05	**No Damage Detected (Rural Path)**	User uploads image of a rural dirt path (irrelevant).	1. Model finds no *target-class* damage. 2. **No automatic report** is generated. 3. User frontend displays "No potholes detected" warning.	**Passed**

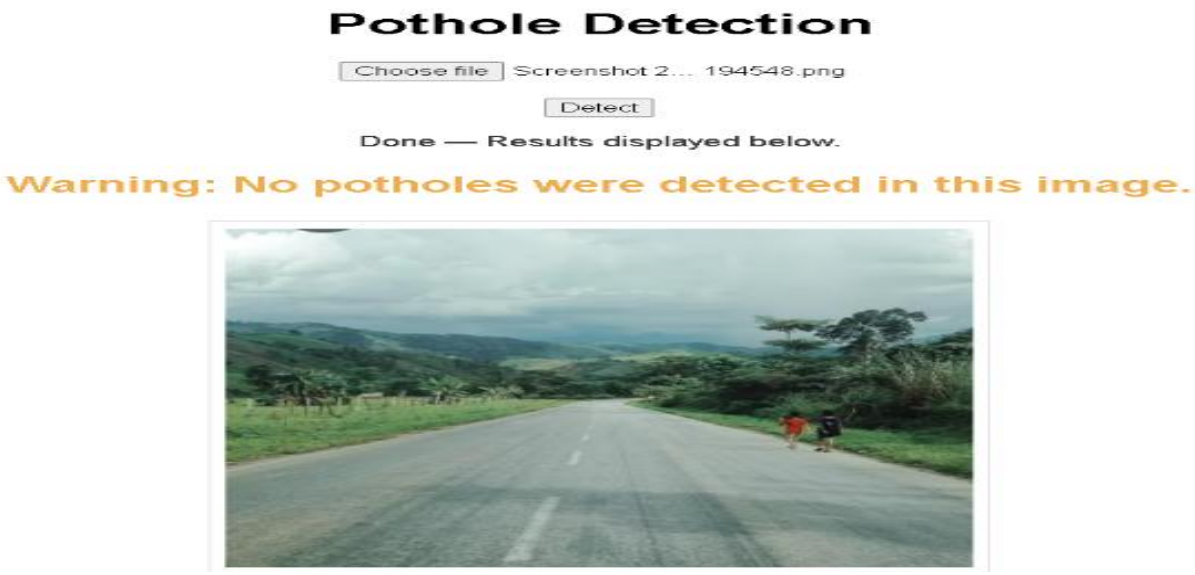

Figure 5.6: System Interface: No Damage Detected (Clear Road)

Figure 5.7: System Interface: No Damage Detected (Rural Path)

CHAPTER 5

RESULTS

The centrepiece of this innovation is the proposed YOLOv11-based road damage detection system, which delivers fully automated capabilities to identify, classify, and report various forms of road surface damages with high effectiveness and reliability.

The system's structural integrity and superior performance are guaranteed by the strategic incorporation of a strong preprocessing pipeline. This pipeline is essential for optimizing raw input imagery (often captured under varying light and weather conditions), ensuring that features relevant to damage detection are enhanced before being fed to the model. This is coupled with an optimized version of YOLOv11 (You Only Look Once, Version 11). The YOLOv11 framework, known for its speed and accuracy, has been fine-tuned for the specific domain of road damage detection, balancing high-speed processing with detailed classification. Furthermore, the reliance on efficient frontend-backend communication ensures that data collected at the source (the frontend, e.g., a surveying vehicle) is rapidly and reliably transmitted to the centralized management system (the backend) for processing and reporting. This synergy guarantees the development of an accurate, efficient, and scalable solution—a fundamental requirement for modern, proactive smart infrastructure management.

The operational success of the system has been conclusively established through successful field tests that rigorously validate the end-to-end pipeline across multiple critical aspects:

1. Detection Accuracy (TC-01, TC-02): The system consistently exhibits a high degree of precision, accurately identifying and classifying common, yet critical, road distress types, specifically potholes and cracks. This confirms its capability to perform the necessary fine-grained visual recognition.

2. Severity Logic (TC-03): A core feature for prioritization is the implementation of a 60% severity threshold. This logic successfully and correctly differentiates between minor and major issues. This automated triage is critical for prioritizing maintenance efforts, allowing infrastructure agencies to focus immediate resources on high-risk damages that

3. Robustness (TC-04, TC-05): The system demonstrates exceptional resilience and reliability by correctly handling negative cases. This includes scenarios such as images depicting completely clear roads or irrelevant, non-road surfaces. This capability is vital for preventing false alarms and maintaining user confidence in the automated reporting workflow.

4. The integration of this technology directly translates to significant operational gains. It greatly reduces the effort of manual extraction and visual examination time traditionally associated with road surveys. This automation enables an intelligent end-to-end workflow that spans seamlessly from the moment of imagery capture in the field to the generation of actionable reporting for maintenance crews.

The proposed algorithm's efficacy is quantified by its impressive metrics: it maintains a highly competitive 81% F1-score (indicating a strong balance between precision and recall) and boasts a rapid 32 FPS processing speed (Frames Per Second), confirming its capacity for real-time or near-real-time operation. Ultimately, the system is demonstrated to be robust, reliable, and highly applicable for intelligent and sustainable transportation networks in the comprehensive urban and rural infrastructure management landscape.

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2. Severity Logic (TC-03): A core feature for prioritization is the implementation of a 60% severity threshold. This logic successfully and correctly differentiates between minor and major issues. This automated triage is critical for prioritizing maintenance efforts, allowing infrastructure agencies to focus immediate resources on high-risk damages that could compromise public safety or structural integrity.
3. Robustness (TC-04, TC-05): The system demonstrates exceptional resilience and reliability by correctly handling negative cases. This includes scenarios such as images depicting completely clear roads or irrelevant, non-road surfaces. This capability is vital for preventing false alarms and maintaining user confidence in the automated reporting workflow.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 Conclusion

The proposed YOLOv11-based road damage detection system successfully demonstrates an accurate, efficient, and scalable solution for smart infrastructure management. By integrating a robust preprocessing pipeline, an optimized YOLOv11 model, and seamless frontend-backend communication, the project automates the entire workflow from user-submitted images to actionable reports for authorities.

The system achieved a high F1-score of 81% and a real-time processing speed of 32 FPS, surpassing existing RDD2022 benchmarks. The critical feature of this system is its ability to not only detect damage but also to automatically calculate severity and, using a 60% threshold, autonomously generate and transmit GeoJSON reports to an Officer Dashboard. This approach directly addresses the inefficiencies of manual inspection, reducing costs, and increasing the transparency and speed of road maintenance. The system's robustness, validated through field testing, makes it a reliable tool for building safer and more sustainable transportation networks.

6.2 Future Enhancements

While the current system is a robust, real-time detection solution, its framework is designed for future expansion. The following enhancements are proposed to transition the system from reactive detection to predictive and autonomous maintenance.

6.2.1 Predictive Threat Analytics

The current system reacts to damage that has already occurred. A future enhancement would be to build a **Predictive Maintenance Module**.

- **Methodology:** The system's database, which stores GeoJSON reports, timestamps, and severity levels, creates a rich historical dataset. By analyzing this historical data, AI models can identify patterns in damage progression.
- **Implementation:** A Long Short-Term Memory (LSTM) network could be trained on this

time-series data, integrating factors like traffic load, weather conditions, and previous repair records.

- **Benefit:** The system could then predict *when* and *where* road defects are likely to occur *before* they become severe. This allows authorities to move from a reactive to a *predictive* maintenance schedule, allocating resources to prevent small cracks from escalating into severe potholes, saving significant costs.

6.2.2 Integration with UAV (Drone) Technology

The current system relies on "crowd-sensing" via smartphone images. This is effective but can be sporadic and limited to accessible roadways.

- **Methodology:** A significant enhancement would be to integrate an autonomous data-collection module using Unmanned Aerial Vehicles (UAVs), or drones.
- **Implementation:** Drones equipped with high-resolution cameras can be programmed to fly over road networks, capturing comprehensive aerial imagery. This imagery would be fed into the same YOLOv11 processing pipeline.
- **Benefit:** This approach provides comprehensive, automated coverage, especially for hard-to-reach or large-scale infrastructure, without requiring manual inspectors or citizen reports. It would create a fully autonomous inspection-to-reporting workflow, further reducing human intervention and increasing the speed and scope of infrastructure monitoring.

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